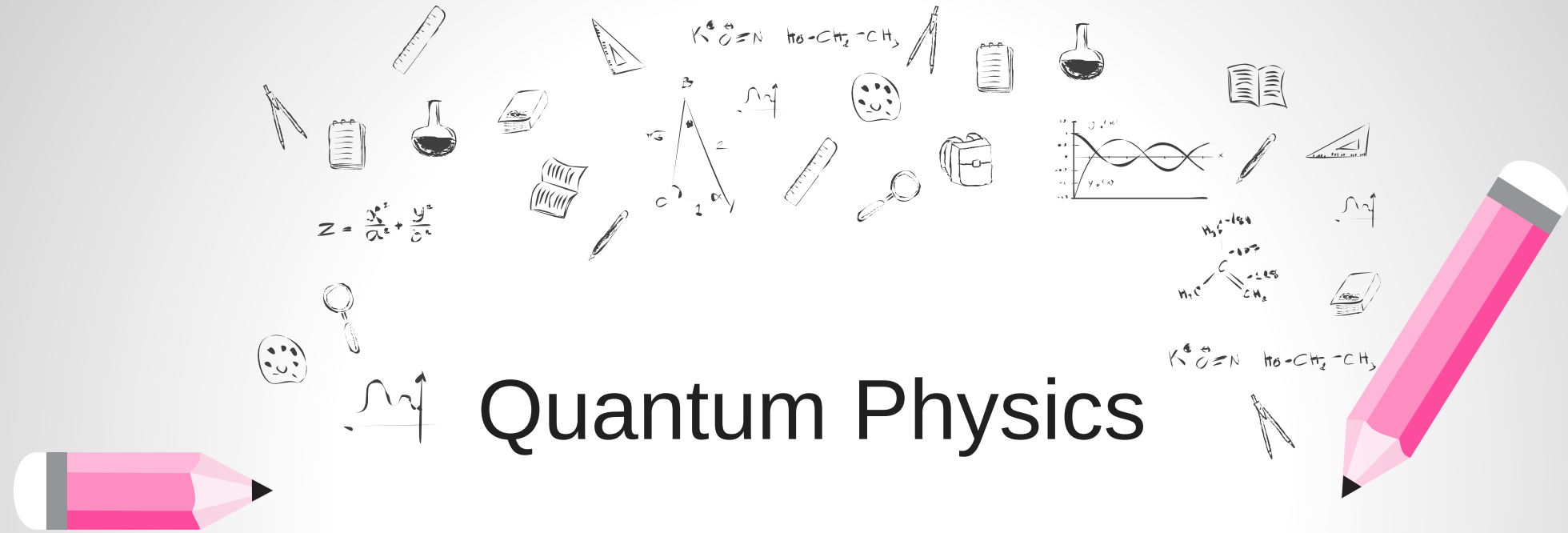




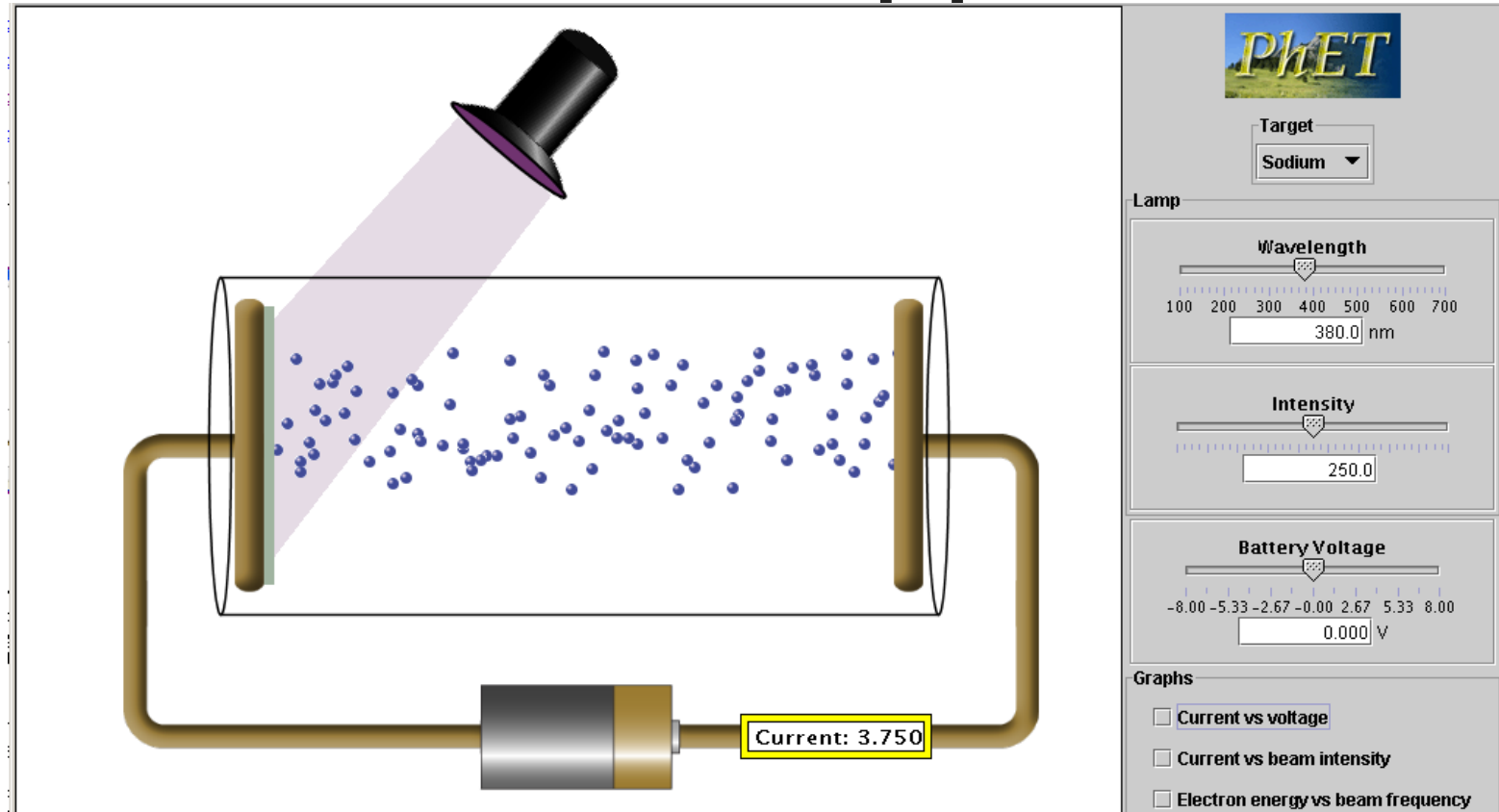
2020

Photoelectric Cell Experiment

2020

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Photoelectric cell Apparatus



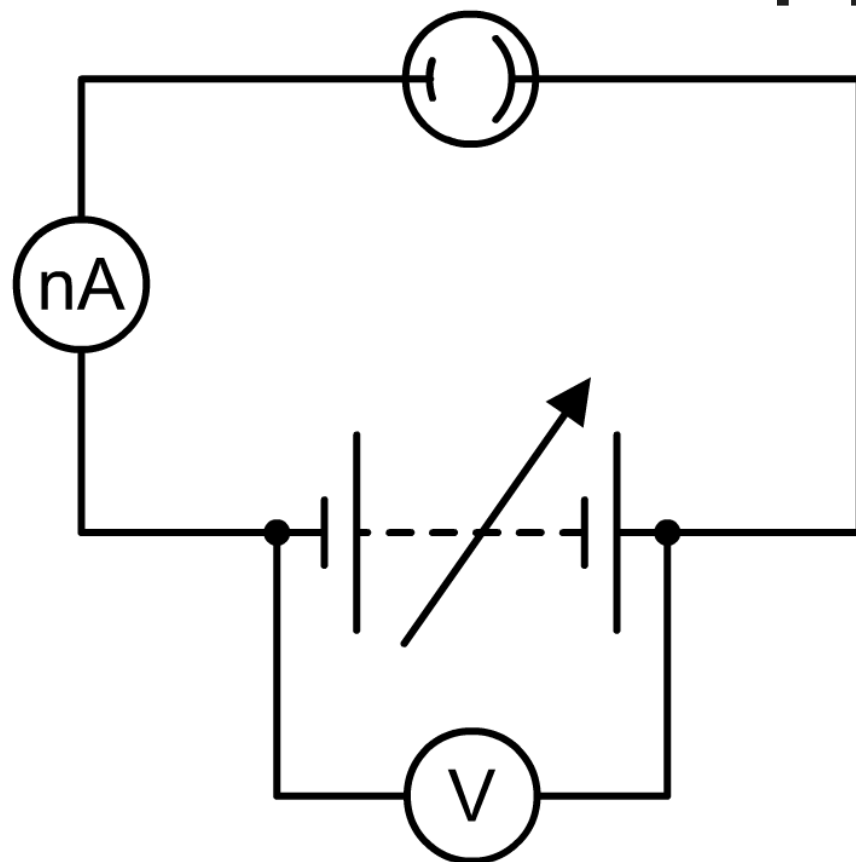
- ▶ Two metal plates in vacuum, adjustable voltage between them, shine light on one plate.
- ▶ Measure current between plates.

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Photoelectric cell Apparatus



- ▶ When light is shone on the cathode of the photocell, electrons are released. They are attracted to the anode, causing a current to flow.

Variables

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- ▶ Frequency of the incident radiation
- ▶ The voltmeter reading – (This represents the maximum KE of the electrons)

$$hf = \phi + \frac{1}{2}mv^2$$

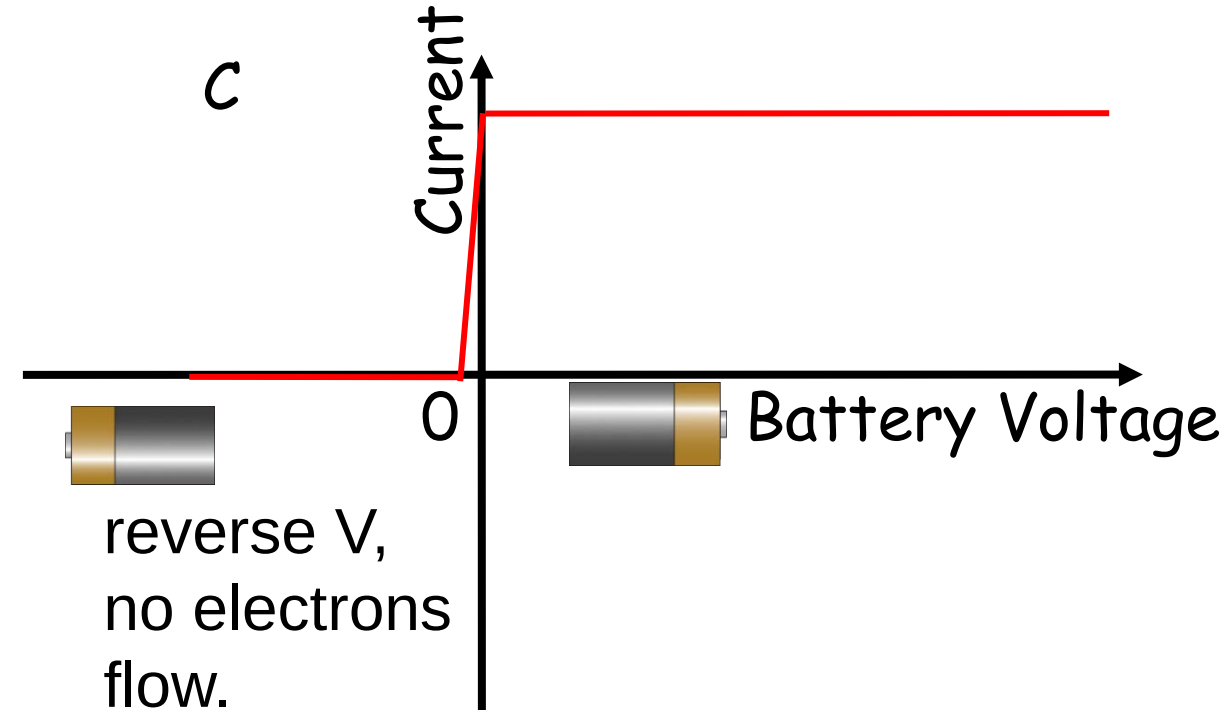
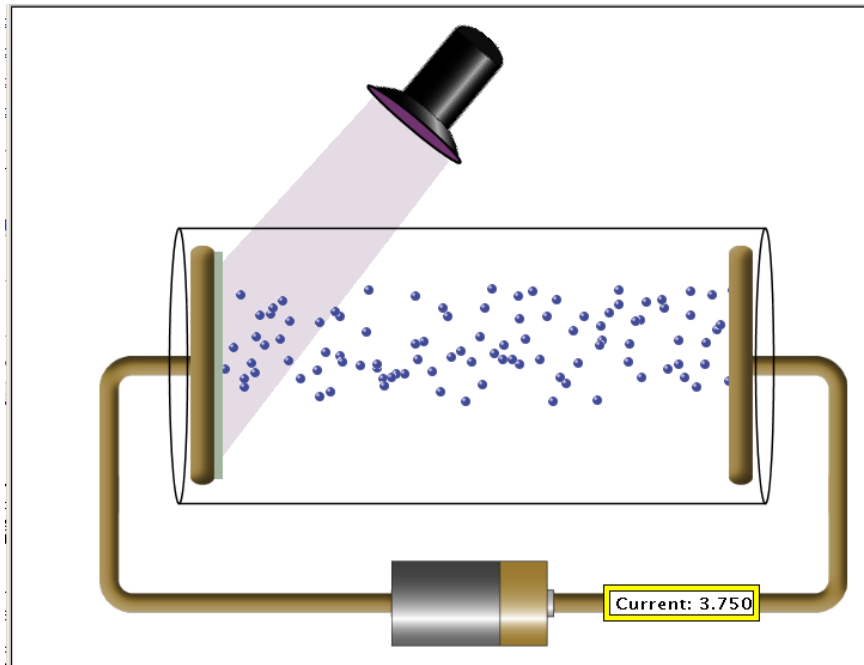
At the Stopping potential, V_s , the KE of the photo electron is = to the max KE



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Photoelectric cell Experiment



What's happening here?

Each electron that is emitted it is accelerated more so hits far plate with higher velocity,

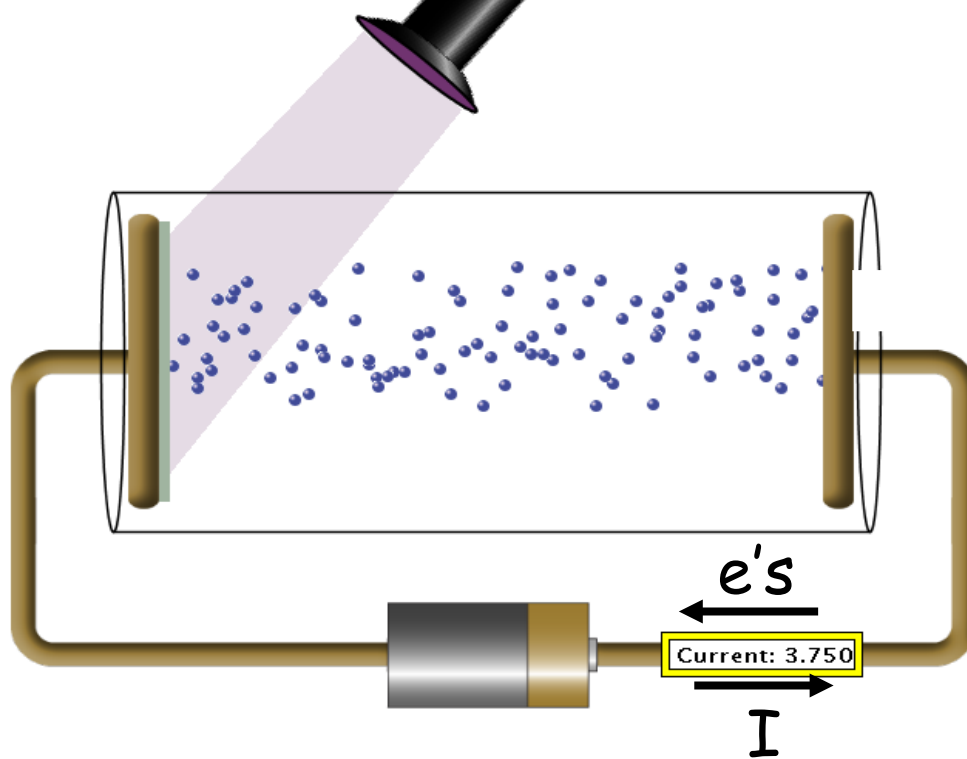
BUT number of electrons = constant
sec

So current is constant!



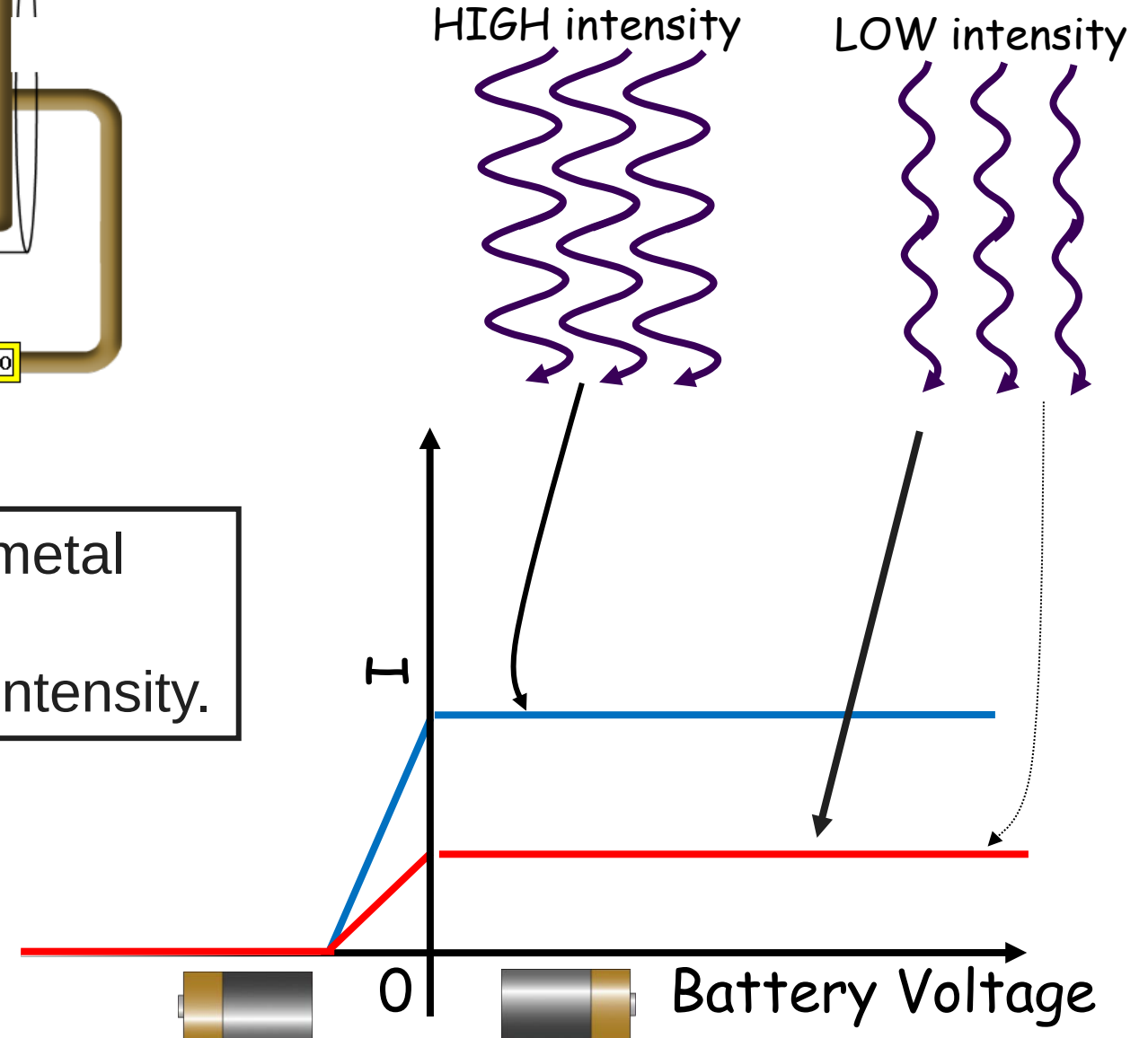
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Fewer electrons emitted off metal
Current decreases.
Current proportional to light intensity.

Effect of Intensity



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Increasing Intensity

- ▶ When intensity increases the number of photons per second increases
- ▶ More photons means more photoelectrons emitted
- ▶ Therefore the ammeter reading increases



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(c) An experiment was carried out using this circuit. The potential difference applied to the photocell opposes the movement of the electrons through the photocell, reducing the current.

- (i) The variable power supply was initially set at 0 V and the reading on the ammeter was $2.4 \mu\text{A}$. The intensity of the ultraviolet light was then increased without changing the frequency. Explain what happened to the reading on the ammeter.

(3)

- (ii) The variable power supply was adjusted until the reading on the ammeter became $0 \mu\text{A}$. The reading on the voltmeter was 3.59 V.

Calculate the work function, in joules, of the metal used in the photocell.

frequency of ultraviolet used = $2.00 \times 10^{15} \text{ Hz}$

(4)



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Electron Diffraction and Interference

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Diffraction of Electrons

- ▶ Observed when
 - ▶ Electrons are made to pass through a gap
 - ▶ When electrons pass through a diffraction grating (such as a thin sheet of graphite)

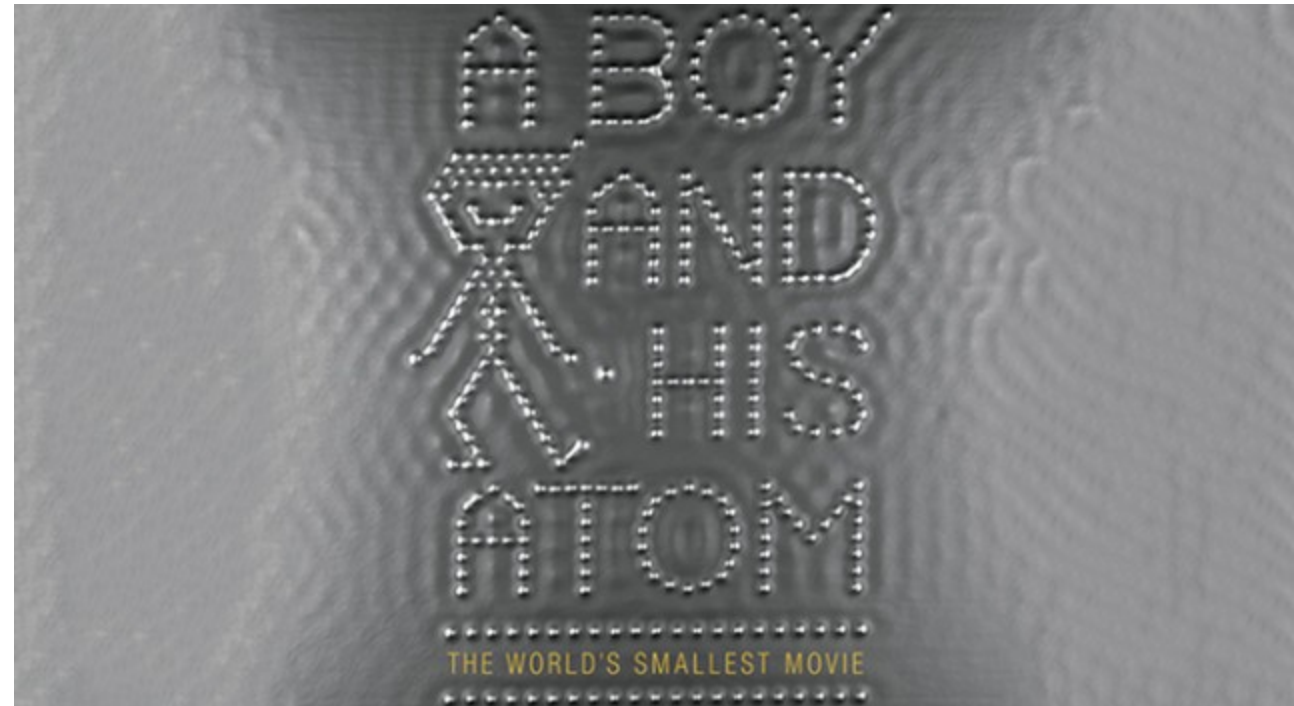


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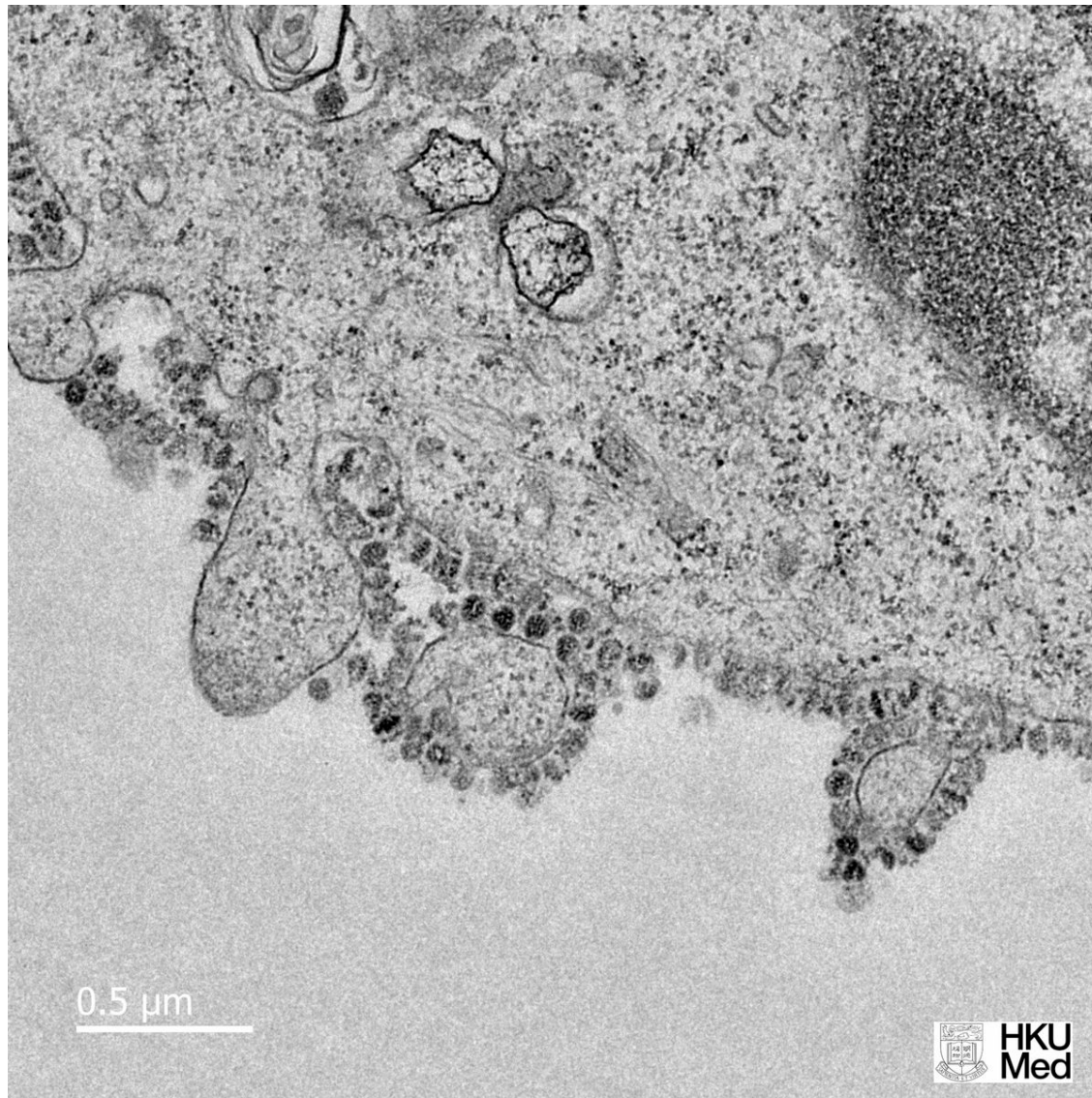
Applications

- ▶ Electron microscope
 - ▶ Since the speed of the electron beam can be controlled by altering the p.d used to accelerate the electrons
 - ▶ We can reduce λ be in the range of 10^{-10}m which is the size of an atom



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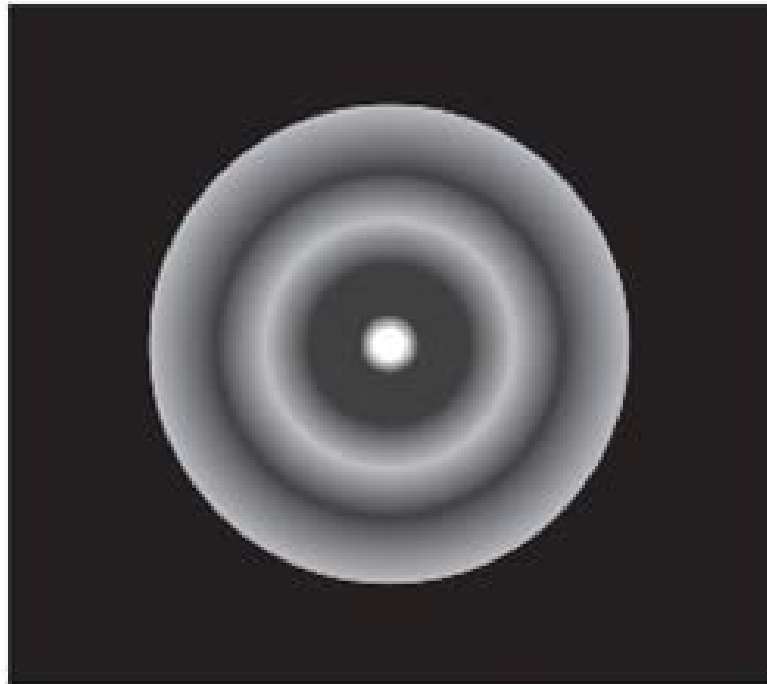
The de Broglie equation

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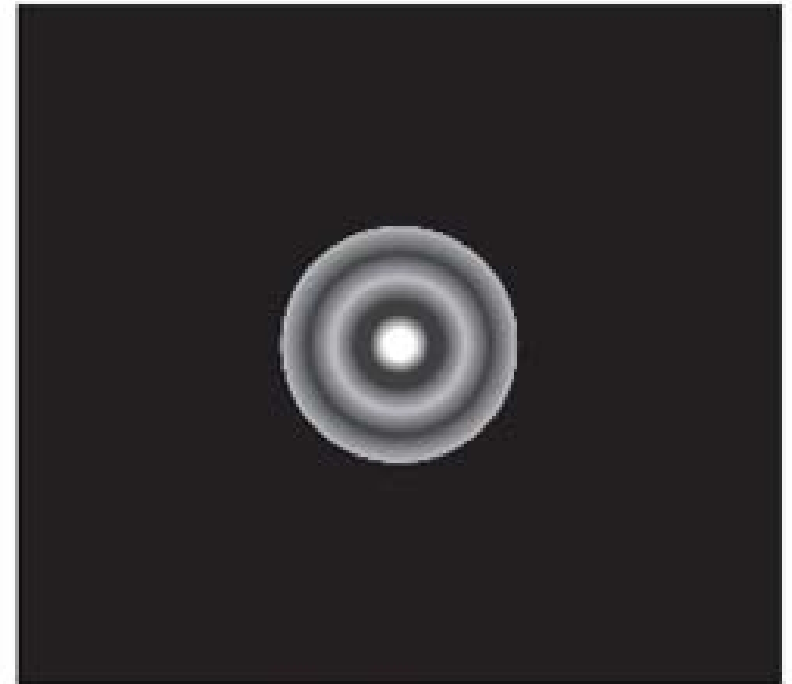
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$$\lambda = \frac{h}{p}$$

- 🎥 This equation explains how the speed of the electron beam affects its wavelength



Slower speed



Faster Speed

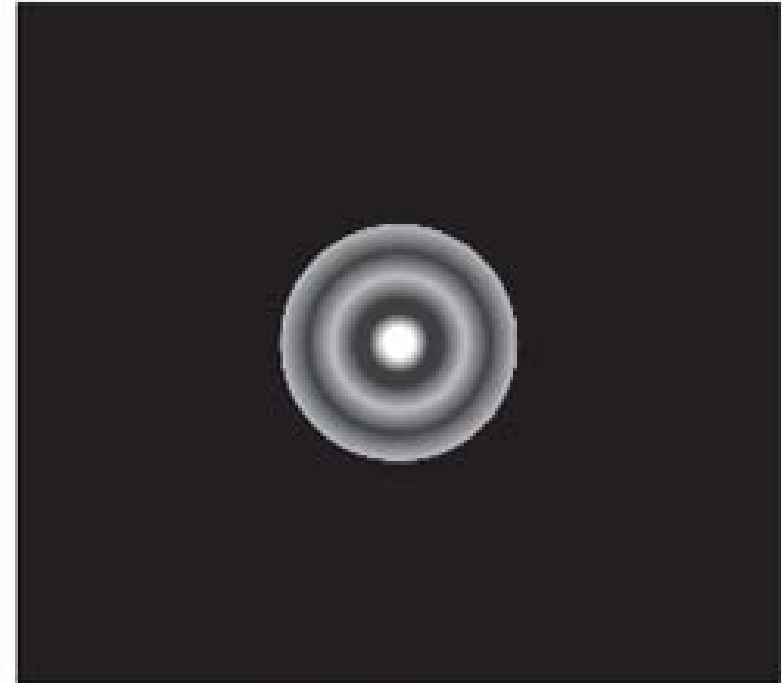
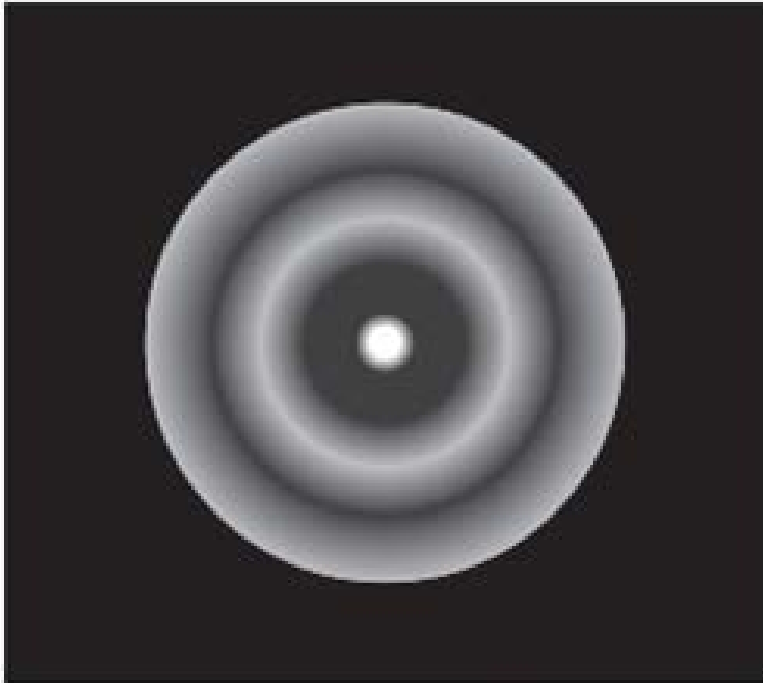


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An electron beam is directed onto crystalline graphite. A fluorescent screen on the other side of the crystal shows the pattern in Figure 1. The brighter areas correspond to higher electron intensity.

The speed of the electrons is increased and the resulting pattern is shown in Figure 2.



* Discuss the conclusions that can be drawn from this information about the behaviour of electrons and the structure of graphite.



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(a) Solar sails are a form of propulsion for spacecraft. The sail is made of a thin sheet of reflective material. When photons of light from the Sun reflect from the material a force is exerted on the sail. The photons reflect with a momentum equal to their initial momentum but in the opposite direction.

(i) Show that a single photon of frequency 1.5×10^{15} Hz has a momentum of about 3×10^{-27} N s.

(2)

.....

.....

.....

.....

(ii) Hence determine the momentum transferred to the solar sail by this photon.

(1)

.....

Momentum transferred =





Thank You